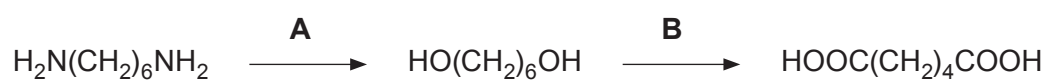


7. Give the displayed formula of the organic compound obtained when cyclohexanol ($\text{C}_6\text{H}_{11}\text{OH}$) reacts with ethanoyl chloride. [1]

8. State the reagents necessary to convert hexane-1,6-diamine to hexane-1,6-dioic acid in two stages. [2]

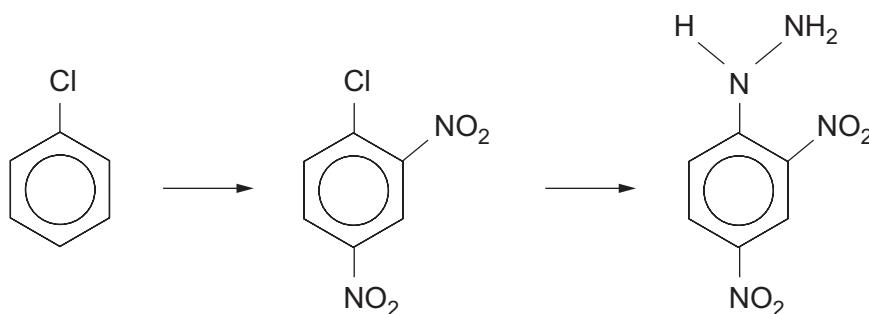


A

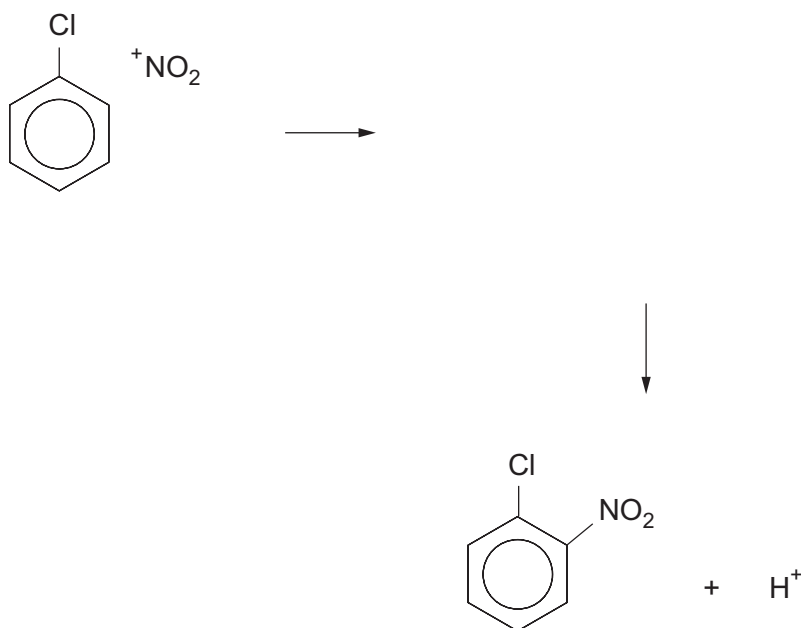
B



7. (a) 2,4-Dinitrophenylhydrazine (2,4-DNPH) can be made by the following route.



- (i) Complete the mechanism for this reaction, which shows the nitration of chlorobenzene. You should use curly arrows and also show the structure of the intermediate. [2]



- (ii) State the type of mechanism occurring in this nitration. [1]

- (b) State the feature of 2,4-DNPH that enables it to react as a base. [1]



(c) 2,4-DNPH can be used as a reagent to identify aldehydes and ketones.

(i) State what is seen when 2,4-DNPH is used to identify an aldehyde or a ketone. [1]

.....

(ii) Explain why 2,4-DNPH is an appropriate reagent to use in their identification. [1]

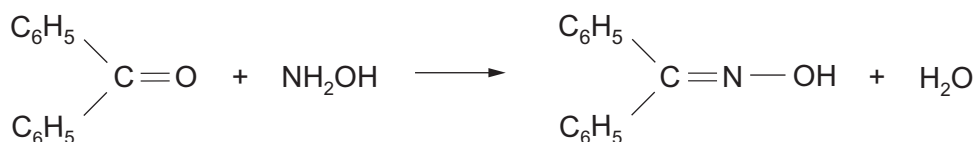
.....

.....

.....

(d) Alternative reagents can be used for this identification. One of them is hydroxylamine, NH_2OH , which produces a white crystalline solid when it reacts with an aldehyde or ketone.

For example with diphenylmethanone



diphenylmethanone oxime
melting temperature 143°C

In an experiment, impure diphenylmethanone oxime was recrystallised from flammable methanol (boiling temperature 65°C).

Outline a **safe** procedure to obtain pure dry crystals of diphenylmethanone oxime using methanol as the solvent. You should assume that all the material present in the crude oxime is soluble in methanol. [5]

.....

.....

.....

.....

.....

.....

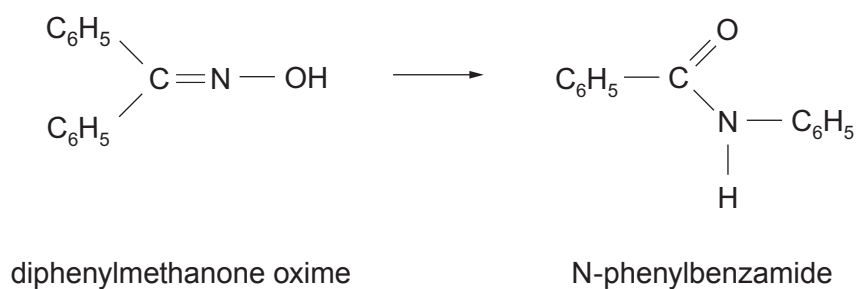
.....

.....

.....



(e) Under suitable conditions diphenylmethanone oxime undergoes a rearrangement.



Describe how the infrared absorption spectra of these two compounds would differ, giving the relevant bonds and their wavenumbers. You should comment on **both** compounds when considering differences. [3]

.....

.....

.....

.....

.....



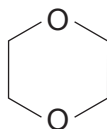
7. Three isomers have the molecular formula $C_4H_8O_2$.

(a) (i) Give the structure of an isomer that gives bubbles of carbon dioxide when tested with aqueous sodium hydrogencarbonate. [1]

(ii) Give the structure of an isomer that is both a ketone and a secondary alcohol. [1]

(b) Describe and explain the high resolution 1H NMR spectrum of 1,4-dioxane.

The position of the signal(s) is **not** required. [1]

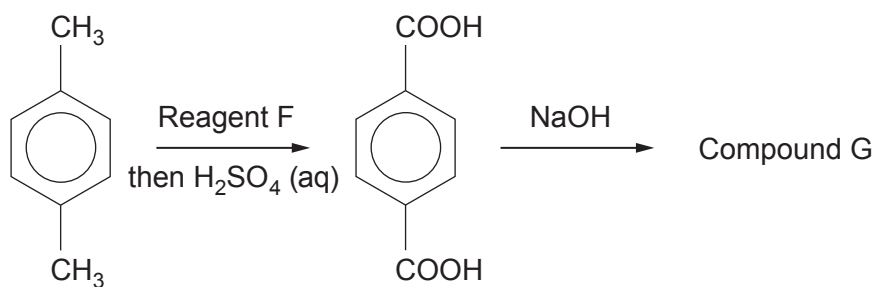


1,4-dioxane

.....
.....



7. Study the reaction sequence below and answer the **two** questions that follow.



(a) Reagent F is

[1]

(b) Give the structure of Compound G.

[1]

